

# KARTIK MISHRA

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## SUMMARY

Machine learning engineer focused on real-time voice AI and low-latency inference at production telecom scale. At Airtel, I own the ASR, model-serving, and async orchestration behind a live broadband voice agent on Airtel IQ, alongside the data platform powering network-health KPIs across 2G/4G/5G. Prior data-engineering work at Moody's (Dagster + weather intelligence) and Verisk (Python automation on AWS SageMaker).

## EDUCATION

**Amity University, Noida, India**

**Aug 2017 – Mar 2021**

**B.Tech. Biotechnology**

**CGPA: 8.15/10**

## WORK EXPERIENCE

### BHARTI AIRTEL

**Gurugram, India**

**Machine Learning Engineer**

**Aug 2025 – Present**

- Built a production real-time voice agent for broadband field diagnostics, hitting sub-1,000 ms round-trip latency across a full STT->LLM->TTS turn-taking loop orchestrated in LangGraph; live on Airtel IQ telephony handling concurrent calls.
- Fine-tuned NVIDIA Nemotron ASR (FastConformer CTC/RNN-T) on IndicVoices and NPTEL, cutting WER 35%->19% (Hindi) and 40%->14% (English), with code-switching handled in a single model.
- Fine-tuned Meta's Omnilingual ASR (1B/3B LLM variants) and Whisper-Hindi2Hinglish-Apex for Hindi/Hinglish telephony to 18.9% WER / 9.0% UER, via a  $\mu$ -law + noise-augmentation pipeline injecting real noise profiles mined from in-house call recordings into clean training data.
- Owned the ASR serving stack end to end: sherpa-onnx export, dual-lane inference on Triton and vLLM, with Silero VAD segmentation, DeepFilterNet suppression, and bandwidth extension to close a confirmed 8->16 kHz telephony audio ceiling.
- Designed the async serving layer: FastAPI microservices, WebSocket audio streaming, and a MongoDB-backed priority queue and call state machine managing concurrent live sessions.
- Rebuilt the telecom data platform (2G/4G/5G) on Docker-based Airflow with CI/CD, orchestrating 20+ PySpark/pandas ETL workflows over terabyte-scale daily broadband data - 75% faster, 85% lower compute cost.
- Leading nationwide Duct operations across 12+ states - a business-critical program directly affecting customer network performance, traffic, and tower/cell operations - as tech lead and primary SME/POC coordinating cross-functional teams India-wide; deployed a real-time monitoring dashboard that eliminated manual inspections and gave stakeholders instant operational visibility.
- Built large-scale PySpark pipelines processing terabytes of broadband customer data daily across India, generating network-health KPIs, churn signals, and CX metrics for Airtel's flagship Broadband Experience initiative.

### MOODY'S INSURANCE SOLUTIONS

**Noida, India**

**Associate Analytics, Model Analyst**

**Apr 2023 – Aug 2025**

- Architected an end-to-end data orchestration framework on Dagster, integrating high-frequency weather data from 30+ global sources with CI/CD automation via Jenkins and GitHub Actions, improving pipeline efficiency by 90%.
- Implemented a Django-based data visualization dashboard using Mapbox, JavaScript, MongoDB, and Python, enabling global wind station tracking and cross-referencing with the Postgres database to identify over 18,000 missing stations
- Collaborated with the Event Response team to automate Support Center uploads via an API, reducing manual upload time from 40–60 minutes to zero, enhancing efficiency for natural disaster reports and forecasts
- Unlocked Visibility into HWind processes by building insightful statistics dashboard using JavaScript, Python, and Django, tracking daily and weekly data feed counts with interactive charts, identifying any problematic data transactions and feed downtimes.

### VERISK ANALYTICS (WOOD MACKENZIE)

**Gurugram, India**

**Data Associate**

**Nov 2021 – Apr 2023**

- Enhanced data collection and processing pipelines in Python on AWS SageMaker, improving efficiency by 60%
- Built Python automation scripts to extract oil price data from country reports, update Excel deliverables, and trigger email notifications on completion, reducing end-to-end processing time by 90%.
- Led project handover sessions with different teams, managed project roadmaps, story allocation, and task division using JIRA
- Crafted a Power BI dashboard to monitor JIRA tickets and track completion statistics, fostering team accountability and ensuring timely task fulfillment; implemented JIRA tickets for day-to-day tasks assigned to individual team members

## PUBLICATIONS

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- Identification of vaccine targets & design of vaccine against SARS-CoV-2 coronavirus using computational and deep learning-based approaches - PeerJ, May 2022
- Identification of vaccine targets in pathogens and design of a vaccine using computational approaches - Nature Scientific Reports, September 2021
- To Study the Effect of Unconventional Treatment Protocol on COVID-19 Patients in Delhi Using Artificial Intelligence Based Methods - SSRN, September 2020

## EXTRACURRICULAR ACTIVITIES

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**Member & AVP Education, Toastmasters Club, West Delhi** - Organized weekly meetings, speaker selection, and onboarded new members (Apr 2023 – Mar 2025)

**Volunteer & Drive Lead, Robin Hood Army, Delhi** - Led 30+ food distribution drives and taught 25+ students weekly (Oct 2022 – Present)

**Coordinator, CRC Placement Cell, Amity University** - Orchestrated 100+ on-campus placement drives and pre-placement prep sessions (Feb 2019 – Mar 2021)

## SKILLS

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**Languages:** Python, SQL

**Speech & ASR:** NVIDIA NeMo, Whisper, Meta Omnilingual ASR (1B/3B), Nemotron/FastConformer, sherpa-onnx, Silero VAD, DeepFilterNet

**ML/AI:** PyTorch, Transformers, HuggingFace, PEFT/LoRA, LangGraph, LangChain

**Serving & Inference:** vLLM, NVIDIA Triton, ONNX Runtime, FastAPI, WebSockets, Docker, CUDA

**Data & Orchestration:** PySpark, Pandas, Delta Lake, Airflow, Dagster, Jenkins, GitHub Actions

**Databases:** PostgreSQL, MongoDB

**Other:** Git, Linux, AWS SageMaker, Streamlit